Jinwon Lee

AI Platform & Reliability Engineer

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PROFILE

AI platform and reliability engineer with about eight years of experience delivering governed agent applications, cloud platforms, and user-facing services into production.

Builds and operates systems across agent runtime integration, delivery automation, access control, observability, incident response, and production stabilization.

SELECTED EVIDENCE

- AWS EKS platform for governed, self-service AX application delivery
- Production reliability for an enterprise AI service used by 30,000 employees
- Korea and UK launches, incident response, and stabilization for a B2C generative AI service

CORE SKILLS

Agent Platform	AWS EKS, agent runtime integration, self-service delivery, automatic domains, existing LLM Gateway API integration
Reliability	Grafana, Prometheus, OpenTelemetry, incident response, rollout stabilization, high availability, auto-scaling
Software & Delivery	Python, Bash, Linux, Kubernetes Operators, ArgoCD, GitHub Actions, GitOps, Helm/Helmfile
Security & Governance	Authentication, authorization, platform security controls, policy enforcement, enterprise governance

EXPERIENCE

Samsung Electronics, AX Development Group

Seoul, South Korea
SRE / AI Platform Engineer
Jan. 2026 - Present

- Design and build an internal AX Agent Platform on AWS EKS, adapting the agent runtime and deployment path to enterprise platform standards.
- Implemented self-service AX application delivery with automatic domain issuance, authorized-only access, and platform controls for authentication, authorization, security, and governance.
- Enabled AX applications to call the existing enterprise LLM Gateway API through platform-side client compatibility and runtime integration.
- Supported a Microsoft-collaborative B2C generative AI service through Korea and UK launch readiness, initial production operations, incident response, and stabilization from Jan. to Jun. 2026.

Samsung Research

Seoul, South Korea
SRE / AI Platform Engineer
Jul. 2024 - Dec. 2025

- Led production delivery and application reliability for a company-wide AI chatbot serving 30,000 employees on internal Kubernetes clusters.
- Owned releases across development, staging, and production while integrating application delivery with security and operational requirements.
- Built container delivery pipelines with Docker, Kubernetes, GitHub Actions, ArgoCD, Helm, and Helmfile.
- Deployed Grafana, Prometheus, and OpenTelemetry observability for shared service visibility and operational diagnosis.

Samsung Electronics, Network Division

Seoul, South Korea
SRE & Software Engineer
Jan. 2021 - Jun. 2024

- Designed and delivered containerized applications with Docker, Helm, Kubernetes Operators, and AWS services including EKS, ECS, Route 53, RDS, and IAM.
- Built and managed OpenShift, Kubernetes, and EKS environments for reliability, high availability, performance, and auto-scaling.
- Designed and developed a Bash-based one-click Kubernetes cluster provisioning system for cloud developers.
- Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.

TmaxOS & TmaxCloud

Gyeonggi, South Korea
Software & Platform Engineer
Jan. 2019 - Dec. 2020

- Built a Siri-like AI assistant on TmaxOS, covering model inference, data preprocessing, system integration, and asynchronous application flows.
- Designed and developed Kubernetes CI/CD and registry operators and deployed a private container image registry.

SELECTED ENGINEERING WORK

Siri-like AI Assistant on TmaxOS | TmaxOS, 2019 - 2020
Built model inference, data preprocessing, system integration, and asynchronous application flows for an OS-level AI assistant.

High-throughput Packet Mirroring | Samsung Electronics Network Division, 2021 - 2024
Developed a Linux application that mirrors and forwards up to 130,000 packets per second under constrained system resources.

Company-wide AI Chatbot | Samsung Research, 2024 - 2025
Led production delivery and reliability for an AI chatbot used by 30,000 employees, with development, staging, and production Kubernetes releases tied to security requirements and Grafana, Prometheus, and OpenTelemetry observability.

B2C Generative AI Service, Korea & UK | Samsung Electronics, Jan. - Jun. 2026
Ran Korea and UK launch readiness, initial production operations, incident response, and stabilization for a Microsoft-collaborative consumer AI service on Azure.

Internal AX Agent Platform | Samsung Electronics, 2026 - Present
Built an AWS EKS platform for governed AX application delivery with runtime adaptation, automatic domains, authorized-only access, and integration with an existing enterprise LLM Gateway API.

LLM-Assisted Cluster Monitoring | Personal project
Built a monitoring system for internal clusters that detects errors, uses an LLM to analyze issues, and emails problem summaries, analysis, and suggested remediation steps.

OfferNow | Personal project, Sep. 2026 - Present
Building a CV-based job recommendation service for Korean IT roles: it turns PDF/DOCX CVs into evidence-linked profiles, collects official postings from 50+ companies while honoring robots.txt, and shows the CV evidence behind every match. Expo (web, iOS, Android), FastAPI, PostgreSQL with row-level security, and provider-neutral LLM calls with per-user limits and cost logging, deployed with Docker Compose and nginx.

AWARDS & ACTIVITIES

Finalist, SK hynix AI Hackathon 2026 Nov. 2026
Advanced through two online qualifying rounds to the offline finals (individual entry).

EDUCATION & RESEARCH

UNIST | M.S. in Computer Science and Engineering Mar. 2017 - Feb. 2019
Master's thesis: Convergence Aware CNN Training | Advisors: Jiwon Seo and Sam H. Noh

UNIST | B.S. in Computer Science and Engineering, Minor in Mathematics Mar. 2012 - Feb. 2017
Undergraduate research: distributed systems for AI model training with TensorFlow

PUBLICATIONS & LANGUAGE

- USENIX NSDI 2019 poster: Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement.
- arXiv 2019: Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement.
- ACM EuroSys 2018 poster: Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism.
- OPIc IH (Intermediate High), Dec. 2025.